

TrES-5b

R-0967

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230728 · created 2026-07-29T16:55:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460153.8385 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.053
Transit depth (Rp/Rs)^2	3.32 +/- 1.94 [%]
Orbital Inclination (inc)	88.31 +/- 2.83
Airmass coefficient 1 (a1)	0.191 +/- 0.092
Airmass coefficient 2 (a2)	1.35 +/- 1.33
Scatter in the residuals of the lightcurve fit is	45.72 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.0862 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.053 vs 0.142 (deviation 0.57 σ)
Duration fit vs published	0.0862 d vs 0.0758 d (deviation 0.92 σ)
Mid-time O-C	8.0 min
Depth SNR	2.6
Residual scatter	45.72 %

Light curve

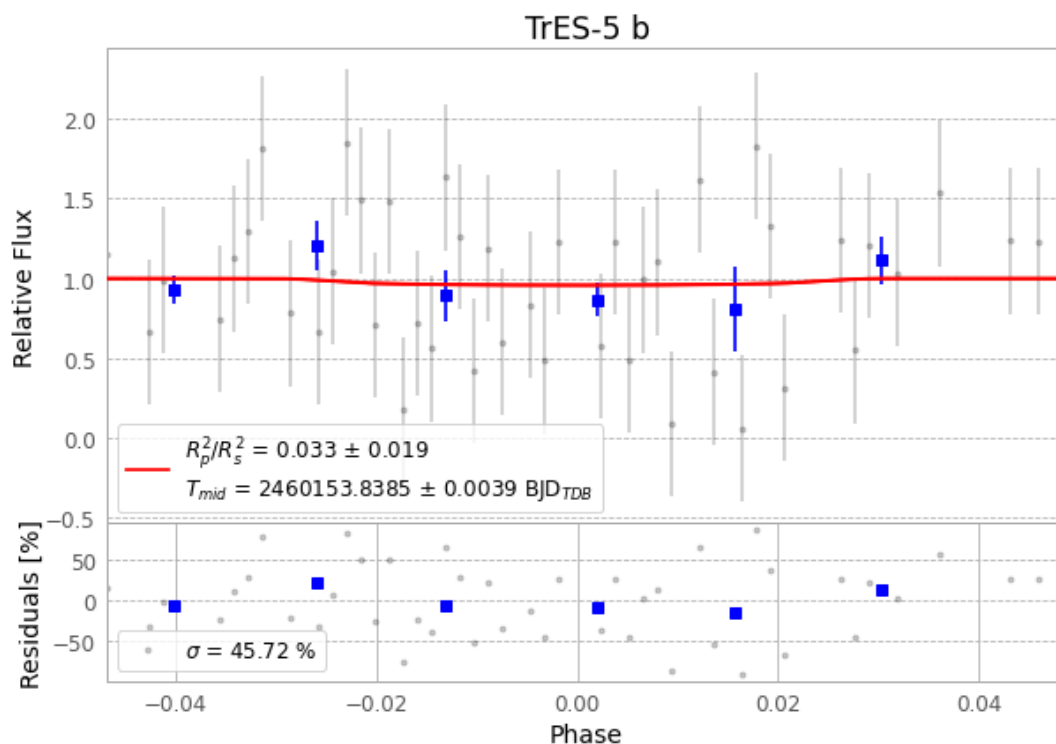


Figure 1 — FinalLightCurve_TrES-5 b_2023-07-27.png (SHA-256 c3069217cc5e5010...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 117], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a870d7d7f6908166...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

64 FITS frames (42 MB), TRES-5230728062413.FITS → TRES-5230728094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230728.log (2163f6f7b0c6...), FinalLightCurve_TrES-5 b_2023-07-27.png (c3069217cc5e...), FinalLightCurve_TrES-5 b_2023-07-27.csv (f5415084a5e1...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:55Z.